

Advanced (Carolina Reaper)

- Q1.** A bookshelf has 5 shelves, and each shelf can hold 3 rows of books. If the bookshelf is currently empty, what is the total number of books it can hold?
(A) 15
(B) 20
(C) 30
(D) 40
(E) 50
- Q2.** A water tank can hold 240 gallons of water. If $\frac{1}{3}$ of the tank is already filled, how many more gallons can be added?
(A) 120
(B) 160
(C) 180
(D) 200
(E) 220
- Q3.** A bakery sells a total of 250 loaves of bread per day. They sell a combination of whole wheat and white bread. If they sell 30 more whole wheat loaves than white bread loaves, what is the number of whole wheat loaves sold?
(A) 125
(B) 130
(C) 135
(D) 140
(E) 145
- Q4.** What is the value of x in the equation $2x + 5 = 11$?
(A) 2
(B) 3
(C) 4
(D) 5
(E) 6
- Q5.** A group of friends want to share some candy equally. If they have 48 pieces of candy and there are 8 friends, how many pieces will each friend get?
(A) 4
(B) 5
(C) 6
(D) 7
(E) 8
- Q6.** A bicycle travels 25 miles in 2 hours. How many miles will it travel in 5 hours?
(A) 50
(B) 60
(C) 62.5
(D) 70
(E) 75
- Q7.** What is the value of y in the equation $y - 3 = 7$?
(A) 5
(B) 6
(C) 8
(D) 10
(E) 12
- Q8.** A clock shows 9:45. What is the next hour and minute that the clock will show the same number of minutes past the hour?
(A) 10:45
(B) 11:45
(C) 12:45
(D) 1:45
(E) 2:45

Open Ended Questions (FRQ)

- Q9.** Tom has been saving money for a new bike and has \$120 in his piggy bank. He wants to buy a bike that costs \$180. If he saves \$5 per week from his allowance, how many weeks will it take him to have enough money to buy the bike?

- Q10.** A company is making a new product that requires 3 parts to be assembled. The parts are made on two different machines: Machine A and Machine B. Machine A makes 2 parts in 5 minutes, and Machine B makes 1 part in 3 minutes. If the company needs to make 24 parts for the new product, what is the minimum amount of time required to make all the parts?

Chef's Tips

- Ensure students understand the difference between natural, whole, and integer numbers.
- Make sure students can identify and generate even and odd numbers.
- Encourage students to recognize and extend numerical patterns in real-life scenarios.